

The empirical recurrence for OEIS A236274: recovering a conjecture that is not written down

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Abstract

OEIS A236274 records “Empirical recurrence of order 65”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 69$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 65, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 65 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236274 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with the upper median unequal to the lower median in every 2×2 subblock.”. It has offset 1 and begins

5328, 137142, 3604782, 98027150, 2691633954, 74525190422, 2069437793694, 57582849812178, ...

The entry states:

Empirical recurrence of order 65 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:12:26, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 69$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 69$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 138$ exact terms of the model returns order 65 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{65} c_i a(n-i),$$

c_1	48	c_{18}	−3070614026757248002	c_{35}	−113347292020162802089471360	c_{52}
c_2	−108	c_{19}	2669643488704649663	c_{36}	1028590795729967083462740684	c_{53}
c_3	−23453	c_{20}	68741479585739935081	c_{37}	610524015486707182802377151	c_{54}
c_4	220776	c_{21}	−63044283251152443662	c_{38}	−2560597175958261361321233643	c_{55}
c_5	4797341	c_{22}	−1220514806091909817247	c_{39}	−2104842222888853344699639150	c_{56}
c_6	−63830931	c_{23}	1049260126052196922826	c_{40}	4607733531931493400778990775	c_{57}
c_7	−543259941	c_{24}	17277749304634305625310	c_{41}	4949802007584351927729038348	c_{58}
c_8	9545722525	c_{25}	−12561019131806081624401	c_{42}	−5694732846523147728311759607	c_{59}
c_9	37234231489	c_{26}	−195317358280579265423097	c_{43}	−8025195273457881644599518438	c_{60}
c_{10}	−908470782792	c_{27}	106079476160156316976885	c_{44}	4363261582322421323557537202	c_{61}
c_{11}	−1516631448354	c_{28}	1760018799971861170592126	c_{45}	8880478143334970868139057544	c_{62}
c_{12}	60226105406660	c_{29}	−577101287754907757616474	c_{46}	−1436486142157331503805883172	c_{63}
c_{13}	26279110171124	c_{30}	−12577525401513192843385811	c_{47}	−6554477896857477332427596824	c_{64}
c_{14}	−2927166701277026	c_{31}	1224493017999794036110042	c_{48}	−611562980904339403380743968	c_{65}
c_{15}	808518946656446	c_{32}	70677224922586533694555122	c_{49}	3114682148038026382528321312	
c_{16}	107754330805526409	c_{33}	9663661104995136077979583	c_{50}	864522445389375449437600608	
c_{17}	−72864696844696423	c_{34}	−308498054281398998112257447	c_{51}	−903038332354559618073375424	

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 65, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $65 + 4$ terms removed returns order 65 as well, so $\deg q_{\min} = 65 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $65 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 65 that this sequence satisfies — and that family turns out to have exactly one member.

References

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